KHALIL KRIFI

+216 28 042 451 | Tunis, Tunisia | khalil.krifi@ensi-uma.tn | LinkedIn | GitHub

EXPERIENCE

Graduation Internship: Natural Language Query Interface for Logs & Metrics Feb 2026 – Jun 2026
Cognira, Tunis, Tunisia

- Architected an event-driven platform translating natural language into OpenSearch DSL and Prometheus PromQL, using Scala (Tapir + Pekko HTTP + jOOQ), Python (FastAPI + LangGraph), and Next.js 14 decoupled over Apache Kafka.
- Engineered a schema-aware LLM pipeline on a self-hosted vLLM model (Qwen2.5-Coder-7B-AWQ) with a LangGraph self-correction loop to validate and repair generated queries before execution.
- Implemented a semantic cache (Redis + vector embeddings) to eliminate duplicate LLM calls, alongside multi-tenant RBAC, TOTP MFA, AES-256 API key encryption, and a BFF proxy with httpOnly JWT storage.
- Deployed to Kubernetes using Kustomize and ArgoCD GitOps, executing automated load and spike tests (Locust, K6) monitored via Prometheus and Grafana.

Software Engineer Intern: Fleet Management System (repo) Jul 2025 – Sep 2025
Omnilink, Tunis, Tunisia (remote)

- Designed a modular fleet management backend utilizing Spring Boot services containerized with Docker to enable independent scaling and isolated domain logic.
- Delivered 40+ GraphQL APIs (Apollo and Netflix DGS) exposing vehicle telemetry, driver allocation, and trip data to the frontend application.
- Implemented real-time WebSocket communication pipelines for live vehicle tracking, status streaming, and instantaneous alert distribution.
- Integrated OAuth 2.0 / OpenID Connect with Keycloak for centralized identity management and fine-grained role-based access control.

AI Research / Engineering Intern: Falsified Signature Detection (repo) Feb 2025 – May 2025
Laval University, Quebec City, Canada (remote)

- Led the design and training of binary classification pipelines using transfer learning backbones, optimizing performance via Optuna to achieve 94% accuracy and 93% F1-score on test data.
- Applied automated image augmentations (Albumentations) to increase model robustness and prevent overfitting across diverse document conditions.
- Incorporated Explainable AI (XAI) techniques including SHAP, LIME, and Grad-CAM to generate visual explanations for system predictions.
- Built a modular microservice pipeline separating PyTorch inference, Spring Boot backend APIs, and an Angular interface with Docker containerization.

Software Engineer Intern: Enterprise Content Management (repo) Jun 2024 – Aug 2024
BFI Group, Tunis, Tunisia (on-site)

- Developed an enterprise document management platform handling files up to 500 MB with hierarchical directory storage, metadata tracking, and document versioning.
- Integrated Elasticsearch to index and search across 20,000+ document records, lowering average retrieval latency to under 300 ms.
- Enforced JWT authentication and granular role-based access controls to restrict document operations based on user permissions.

PROJECTS

Etude.AI: Microservice Study Platform Scalable study assistant backend built with Java and Python.

- Built a reactive microservice architecture combining Spring Boot (WebClient) and FastAPI for asynchronous task processing.
- Configured resilient circuit breakers and automatic retry mechanisms with backoff strategies to protect against downstream service failures.
- Enforced API rate limiting using Token Bucket algorithms (Bucket4j for Java and SlowAPI for Python) to protect server bandwidth.
- Deployed services on Kubernetes with automated horizontal pod autoscaling, health monitoring, and error reporting via Prometheus, Grafana, and Sentry.

EDUCATION

Computer Science Engineering, ENSI, University of Manouba Sep 2023 – Jun 2026
Engineering Preparatory Studies, IPEIN, Nabeul Sep 2021 – Jun 2023

SKILLS

Programming Languages: Java, Scala, Go, TypeScript, Python, C/C++, SQL, GraphQL
Technologies: Git, Docker, Kubernetes, Spring Boot, Angular, React, PostgreSQL, Spark, Hadoop, HBase, Power BI
Languages: Arabic (Native), French (Fluent), English (Fluent) **Interests:** Chess, Piano